Samyak Rai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.41
Lotus Valley International School	Class XII (CBSE)	2024	95.60%
Lotus Valley International School	Class X (CBSE)	2022	95.00%

EXPERIENCE

NIUS Research Intern | *Under Prof. Anirbhan Pathak* **[**wikipedia**]** *(JIIT)*

- Conducted research on **fault-tolerant Quantum Error Correction** for **photonic quantum computing systems**, targeting a 6-qubit photonic architecture developed at **IISc Bangalore**. Synthesized findings from recent literature to guide the design of **photonic QEC schemes**.
- Investigated **post-selection-based** error detection, **measurement-free** QEC, and **topological** error correction to identify hardware-efficient fault-tolerant protocols. Developed simulation frameworks such as **Pauli Frame** and **STIM** for high-performance simulation of **quantum stabilizer circuits** for quantum error correction and detection and analyze logical error behavior under **realistic noise models**.

NIUS Physics 22.1 Camp | *HBCSE, Tata Institute of Fundamental Research (TIFR)*

- Selected through a highly competitive **nationwide process** as one of **70 participants** in the National Initiative on Undergraduate Science (NIUS) Physics Program. Engaged in a research-oriented curriculum featuring lectures, laboratory and theory-based problem-solving sessions, and discussions with distinguished physicists representing a broad spectrum of theoretical and experimental research.
- Awarded a **NIUS Research Internship** at JIIT under **Prof. Anirbhan Pathak** in Quantum Error Correction.

PROJECTS

Optimising Lagrangian Neural Networks [Github] | *Python, PyTorch*

- Investigating the effect of neural network architectures and initialization strategies on the performance of **Lagrangian Neural Networks for physics-informed learning**. Benchmarking model stability and generalization across multiple dynamical systems including pendulum and Duffing oscillator datasets. Developing automated evaluation pipelines to compare convergence behavior and predictive accuracy over multiple random seeds.

Gravitational Wave Identification using MCMC [Github] | *Numpy, Pandas, Seaborn, Git*

- Developed a **Bayesian parameter estimation** framework using the **Metropolis-Hastings Markov Chain Monte Carlo** algorithm to recover gravitational wave parameters from **noisy time-series strain data** generated from analytical waveform models. Designed configurable synthetic data generation and **heteroscedastic noise models** to evaluate inference robustness under **varying signal-to-noise ratios** and observational conditions.
- **Optimized proposal distributions** and constructed new likelihood functions to improve chain convergence, acceptance rate, and numerical stability, achieving **near-optimal** MCMC sampling performance.
- Analyzed posterior distributions using **Effective Sample Size (ESS)**, **Monte Carlo Standard Error (MCSE)**, acceptance ratio, covariance analysis, and trace diagnostics to assess sampling quality and parameter uncertainty.

Lyapunov Exponent Analysis of Chaotic Dynamical Systems [Github] | *NumPy, Pytorch, Matplotlib*

- Constructed numerical methods to compute **Lyapunov exponents for nonlinear dynamical systems** including the **Hénon map** using QR-based orthogonalization. Using parameter sweeps to visualize chaotic and stable regions through Lyapunov exponent phase diagrams. Investigating numerical stability, convergence for large-scale simulations.

Analysis of Quantum Walks in Defected Lattices [Github**] |** *Qiskit Aer, Python, Git*

- Developed a comprehensive simulation framework for **Continuous-Time and Discrete-Time Quantum Walks** to investigate **defect induced quantum transport** phenomena and their applications, including localization, scattering, reflection, transmission, and trapping. Furthermore, I also replicated result from multiple research papers on scalable

quantum circuit implementations of cyclic quantum walks on 4-node and 8-node lattices in Qiskit, and showcased higher performance as compared to the literature used as reference.
- Performed **hardware-aware simulations** using calibration data from IBM Quantum (Fez) and quantified Algorithmic robustness through **Hellinger Fidelity** analysis between ideal and noisy probability distributions. Evaluated the impact of circuit depth, multi-controlled gate decompositions, and noise on scalability of quantum walk algorithms, identifying key limitations of NISQ era quantum devices.

CERTIFICATIONS & AWARDS

Academic Excellence Award for 2026

Secured the highest Grade Point Average (GPA) in B.Tech. Engineering Physics for the calendar year 2025.

OCSC Chemistry and INPhO 2024

- Participated in **INChO** and secured a rank among the **top 35 students in India**, qualifying for the **Orientation-cum-Selection Camp (OCSC)** for the International Chemistry Olympiad 2024.
- Selected among the **top 300 students in India** for the **Indian National Physics Olympiad (INPhO) 2024**.

SKILLS

Programming: Python, Rust, C, Julia, LaTeX, Typst, Bash, Neovim, AppleScript, Verilog, Lean 4

Frameworks & Tools: SciPy, Qiskit, PennyLane Quantum, Git, Docker, Linux & Unix, Ollama, Hugo, Quartz 4, Ratatui

Extra-Curricular: *nix ricing, Hybrid Athlete, Bodybuilding, Powerlifting, Swimming, Running, Sketching, Origami.